

## VARIABLE NAMES & EXPLANATIONS FOR MIDUS 3 PROJECT 5

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### **Character 1:** MIDUS Sample

C = MIDUS 3

### **Character 2:** Project #

5 = Project #5

### **Character 3:** Measure

S = Self-reports

B = Startle Eyeblink

C = Corrugator EMG

L = Zygomaticus EMG

K = Heart Rate Variability (QRSTool and CMetX – used in MR1 only)

V = Heart Rate Variability (NeuroKit2 – used in M3)

R = Psychophysiology Emotional Response Task - Response Times

A = Psychophysiology Emotional Response Task - Response Accuracy

N = CANTAB Cognitive measures

D = Cube & Paper Test

F = Free Recall

T = Picture Ratings

P = Participant Characteristics

H = Handedness

O = Hearing Test

I = General MRI Information

E = Extracted Structural Brain Measurements

W = Extracted Diffusion Weighted Imaging Measurements

*Remaining characters differ to build discrete variables, but all within the given measure type set by character 3.*

### **For Ch3 = S** (i.e., Self-reports):

#### **Character(s) 4 (& 5):** Scale

DP = Dispositional Positive Affect Scale (DPES)

P = Positive and Negative Affect Schedule (PANAS)

IR = Interpersonal Reactivity Index (IRI)

S = Spielberger State-Trait Anxiety Scale (STAI)

E = Emotion Regulation Questionnaire (ERQ)

### **For Ch4 & 5 = DP** (i.e., DPES):

#### **Character 6:** Subscale

C = Contentment

J = Joy

H = Hope

L = Love/Attachment

D = Desire

O = Compassion

P = Pride

G = Gratitude

A = Amusement

W = Awe

I = Interest

T = All scales summary score  
**Characters 7 & 8:** Measure  
None = Summary Measures  
Numbers = Individual Questions

**For Ch4 = P** (i.e., PANAS):

**Character 5:** Time Point

1 = PANAS-Now, Time 1 (prior to Psychophysiology Emotional Response Task)  
2 = PANAS-Now, Time 2 (after Psychophysiology Emotional Response Task)  
3 = PANAS-Now, Time 3 (prior to fMRI Emotional Response Task)  
4 = PANAS-Now, Time 4 (after fMRI Emotional Response Task)  
G = PANAS-General

**Character 6:** Subscale

P = Positive Affect  
N = Negative Affect

**Characters 7 & 8:** Measure

None = Summary Measures  
Numbers = Individual Questions

**For Ch4 & 5 = IR** (i.e., IRI):

**Characters 6 & 7:** Subscale

PT = Perspective-Taking Scale  
FS = Fantasy Scale  
EC = Empathic Concern Scale  
PD = Personal Distress Scale  
None = All scales summary score

**Characters 8 & 9:** Measure

None = Summary Measures  
Numbers = Individual Questions

**For Ch4 = S** (i.e., STAI):

**Character 5:** Time Point

1 = STAI-X1, Time 1 (prior to Psychophysiology Emotional Response Task)  
2 = STAI-X1, Time 2 (after Psychophysiology Emotional Response Task)  
3 = STAI-X1, Time 3 (prior to fMRI Emotional Response Task)  
4 = STAI-X1, Time 4 (after fMRI Emotional Response Task)  
T = STAI-X2, Trait anxiety form of STAI

**Characters 6 & 7:** Measure

None = Summary Measures  
Numbers = Individual Questions numbers.

**For Ch4 = E** (i.e., ERQ):

**Character 5:** Subscale

R = Reappraisal  
S = Suppression

**Characters 6 & 7:** Measure

None = Summary Measures  
Numbers = Individual Questions numbers.

**For Ch3 = B** (i.e., Eyeblick Startle):

C5B = Number of valid eyeblick startle responses measured over entire paradigm.

C5BF = Filter for good eyeblick startle response data

**Character 4:** Picture Valence

N = Negative  
O = Neutral  
P = Positive

**Character 5: Probe Time**

E = Early (2900 ms after picture onset)  
M = Mid (4400 ms after picture onset)  
L = Late (5900 ms after picture onset)

**Character 6: Metric**

A = Amplitude (includes only responses, so assesses height of response)  
M = Magnitude (includes no responses as a zero, so averaging will be affected by no responses)

**For Ch3 = C or L (i.e., Corrugator and Zygomaticus EMG):**

C5C = Filter for good corrugator data (bad corrugator data might exhibit high levels of noise and/or artifact)

C5L = Filter for good zygomatic data (bad zygomatic data might exhibit high levels of noise and/or artifact)

**Character 4: Picture Valence**

N = Negative  
O = Neutral  
P = Positive

**Character 5: Time**

E = Early (1-4 seconds following picture onset)  
M = Middle (5-8 seconds following picture onset)  
L = Late (9-12 seconds following picture onset)

**For Ch3 = V (i.e., Heart Rate Variability – processed with NeuroKit2):**

**Character 4: Session Type**

1 = Psychophysiology baseline  
2 = MRI resting state

**Character 5:**

Q = Data Quality Filter  
B = Beats Removed Filter  
M = Total time masked out in seconds  
I = Number of IBI's used  
T = Time measures  
F = Frequency measures  
N = Nonlinear measures

**For Ch5 = I (Number of IBI's used):**

**Character 6:**

T = Total IBIs  
N = IBIs used for Nonlinear stats

**For Ch5 = T (i.e., Time measures):**

**Characters 6 & 7:**

MH = Mean Heart Rate  
MI = Mean Interbeat Interval  
SN = Standard Deviation of RR beats (SDNN)  
RM = Root Mean Squared Successive Differences of RR intervals (RMSSD)  
PN = Percent of RR intervals over 50 milliseconds (pNN50)  
SD = Standard deviation of successive differences between RR intervals (SDSD)

**For Ch5 = F** (i.e., Frequency measures):

**Characters 6 – 9:**

LF = Spectral power of low frequencies (.04-.15 Hz)

HF = Spectral power of high frequencies (.15-.4 Hz)

TP = Total spectral power

LFHF = Ratio obtained by dividing the low frequency power by the high frequency power

LFN = Normalized low frequency, dividing the low frequency power by the total power

HFN = Normalize high frequency, dividing the high frequency power by the total power

LNHF = Log transformed HF

**For Ch5 = N** (i.e., Nonlinear measures):

**Characters 6 – 8:**

CV = Cardioagal index (CVI) (Toichi, 1997)

CS = Cardiosympathetic Index (CSI) (Toichi, 1997)

CA = Contributions of heart rate accelerations to short-term HRV (Piskorski, 2011)

CD = Contributions of heart rate decelerations to short-term HRV (Piskorski, 2011)

SDA = Short term variance of contributions to accelerations (Piskorski, 2011)

SDD = Short-term variance of contributions to decelerations (Piskorski, 2011)

**For Ch3 = R or A** (i.e., Psychophysiology Emotional Response Task reaction time and accuracy):

**Character 4:** Picture Valence

N = Negative

O = Neutral

P = Positive

**For Ch3 = N** (i.e., CANTAB cognitive assessments):

**Character 4:** Test type

M = Motor Screening Task (MOT)

I = Intra-Extra Dimensional Set Shift (IED)

A = Affective Go/No-Go (AGN)

S = Information Sampling Task (IST)

T = Attention Switching Task (AST)

E = Emotion Recognition Task (ERT)

G = Cambridge Gambling Task (CGT)

**For Ch4 = M** (i.e., MOT):

**Character 5:** Measure

E = Mean Error

L = Mean Latency

**For Ch4 = I** (i.e., IED):

**Character(s) 5 (& 6):** Measure type

Numbers = Summary stage-related measures. See table in

*M3\_P5\_DOCUMENTATION\_OF\_CANTAB\_20260910* for list of measures

T = Totals

C = Calculated Measures  
S = Stages

**For Ch5 = T or C:**

**Characters 6 & 7:** Measure

Numbers = Total measures. See table in

*M3\_P5\_DOCUMENTATION\_OF\_CANTAB\_20260910* for list of measures

**For Ch5 = S:**

**Character 6:** Type

T = Trials per stage

E = Errors per stage

**Character 7:** Stage Number

Numbers = Stages #1-9

**For Ch4 = A (i.e., AGN):**

**Character 5:** Measure

R = Affective Response Bias (Mean)

L = Mean Correct Latency

T = Total Incorrect (Commissions/Omissions errors)

**For Ch5 = L:**

**Characters 6 & 7:** Trial Type

Numbers = Condition (Positive/Negative/Neutral, Shift/Non-shift)

**For Ch5 = T:**

**Character 6:** Total Incorrect Responses/Non-responses

M = Total Commissions

O = Total Omissions

**Character 7:** Trial Type

None = Total

Numbers = Condition (Positive/Negative/Neutral, Shift/Non-shift)

**For Ch4 = S (i.e., IST):**

**Characters 5 & 6:** Measure

Numbers = See table in *M3\_P5\_DOCUMENTATION\_OF\_CANTAB\_20260910* for list of measures

**For Ch4 = T (i.e., AST):**

**Character 5:** Measure Type

T = Totals

P = Percentages

L = Latency-Related Measures

C = Cost-Related Measures

**For Ch5 = T, P, or L:**

**Characters 6 & 7:** Measure

Numbers = See table in

*M3\_P5\_DOCUMENTATION\_OF\_CANTAB\_20260910* for list of measures

**For Ch5 = C:**

**Character 6:** Measure/Trial Type

C = Mean Congruency Cost

S = Mean Switch Cost

**For Ch6 = C or S:**

**Character 7:** Response Type

C = Correct

I = Incorrect

None = All Responses (Correct & Incorrect)

**For Ch4 = E (i.e., ERT):**

**Character 5:** Measure Type

P = Percentages

T = Totals

L = Latency-Related Measures

**For Ch5 = P or T:**

**Character 6:** Response Type

C = Correct

I = Incorrect

**Character 7:** Stimulus Type

Numbers = See table in

*M3\_P5\_DOCUMENTATION\_OF\_CANTAB\_20260910* for list  
of measures

None = Total Correct (All Stimulus Types)

**For Ch5 = L:**

**Characters 6 & 7:** Stimulus/Response Type

Numbers = See table in

*M3\_P5\_DOCUMENTATION\_OF\_CANTAB\_20260910* for list  
of measures

None = Mean Overall Response Latency

**For Ch4 = G (i.e., CGT):**

**Character 5:** Measure Type

A = Delay Aversion

D = Deliberation Time

P = Overall Proportion Bet

Q = Quality of Decision-Making

J = Risk Adjustment

R = Risk Taking

**Character 6:** Trial Type

A = Ascending Trials

D = Descending Trials

None = All Trials

**For Ch3 = D (i.e., Cube & Paper Test):**

C5D = Cube & Paper Total Correct

**Character 4:** Measure

R = Cube & Paper Total Number of Responses

C = Cube subset

P = Paper subset

**For Ch4 = C or P:**

**Character 5:** Subset - Correct vs Number of Responses

A = Number of Correct Response

B = Number of Responses

**For Ch3 = F** (i.e., Free Recall):

**Character 4:** Measure

R = Total Recalled

M = Exposed to MRI pictures in addition to psychophysiology task pictures prior to completing free recall

S = Total Recalled (Social)

X = Total Recalled (Non-Social)

P = Total Recalled (Positive)

N = Total Recalled (Negative)

O = Total Recalled (Neutral)

**For Ch3 = T** (i.e., Picture Ratings):

**Character 4:** Rating Scale

V = Valence

A = Arousal

**Character 5:** Picture Valence

P = Positive

N = Negative

O = Neutral

**Character 6:** Session

1 = Psychophysiology

**For Ch3 = P** (i.e., Participant Characteristic):

C5PAGE = Age at P5 visit

C5PDATE\_MO = Month of P5 data collection

C5PDATE\_YR = Year of P5 data collection

**For Ch3 = H** (i.e., Handedness):

C5HAND = Handedness

**For Ch3 = O** (i.e., Hearing Test):

**Character 4:** Side of hearing test or hearing aid use

L = Left Ear

R = Right Ear

A = Hearing aid worn in at least one ear during test

**For Ch4 = L or R:**

**Character 5:** Frequency of tone

1 = 250 Hz

2 = 500 Hz

3 = 1000 Hz

4 = 2000 Hz

5 = 4000 Hz

**For Ch3 = I** (i.e., General MRI Information):

**Character 4:**

C = Filter for participation in MRI imaging protocol (completed at least T1-weighted structural scan)  
F = Radiologist flagged abnormal structural MRI  
N = Self-reported neurological condition  
T = Time MRI started (military time)

**For Ch4 = T:**

**Character 5:** MRI scan  
S = Structural scan start time  
F = Functional scan start time

**For Ch3 = E (i.e., Extracted Structural Brain Measurements):**

**Character 4:** Measurement Type  
A = Cortical Area  
C = Cortical Curvature  
T = Cortical Thickness  
V = Cortical Volume  
S = Subcortical Volume  
B = Brain-Predicted Age

**For Ch4 = A, C, T, V, S:**

**Character 5:** Brain Hemisphere  
L = Left Hemisphere  
R = Right Hemisphere  
N = N/A: Measure is bilateral  
**Character 6:** Freesurfer Brain Atlas or Module  
D = Destrieux  
K = Desikan-Killiany  
T = Desikan-Killiany-Tourville (DKT)  
A = Aseg Subcortical Atlas or Hippocampal Subfield/Amygdala Nuclei Module

**Characters 7 & 8:**

Numbers = See table in  
*M3\_P5\_DOCUMENTATION\_OF\_BRAIN\_MEASURES\_20260910*  
for list of measures

**For Ch4 = B (i.e., Brain-Based Aging Algorithms):**

**Character 5:** Algorithm  
None = C5EB - Cole brainageR v1.0 - <https://github.com/james-cole/brainageR/tree/1.0>  
C = Cole brainageR v2.0 - <https://github.com/james-cole/brainageR/tree/2.0>  
T = TSAN brainage - PMID: 34086565 - <https://github.com/Milan-BUAA/TSAN-brain-age-estimation>  
P = 3D-CNN brainage - PMID: 36595679 - [https://github.com/irimia-laboratory/USC\\_BA\\_estimator](https://github.com/irimia-laboratory/USC_BA_estimator)  
D = DunedinPACNI Dunedin Pace of Aging Calculated from NeuroImaging - PMID: 40595015 - <https://github.com/etw11/DunedinPACNI>

**For Ch3 = W (i.e., Extracted Diffusion Weighted Imaging Measurements):**

**Character 4:** Measurement Type  
F = Fractional Anisotropy (FA)  
M = Mean Diffusivity (MD)  
R = Radial Diffusivity (RD)

A = Axial Diffusivity (AD)  
N = Mean Kurtosis (MK)  
S = Radial Kurtosis (RK)  
B = Axial Kurtosis (AK)  
X = Axonal Water Fraction (AWF)  
I = Intra-axonal diffusivity (ias\_Da)  
P = Extra-axonal radial diffusivity (eas\_de\_perp)  
T = Extra-axonal tortuosity (eas\_tort)  
D = Neurite density index (NDI)  
V = Orientation dispersion index (ODI)  
C = Fraction of isotropic diffusion (FISO or CSF)

**Character 5: Brain Hemisphere**

G = Global Measure  
L = Left Hemisphere  
R = Right Hemisphere  
N = N/A: Measure is bilateral

**For Ch5 = G:**

**Character 6: Tissue type**

None = White Matter  
A = Gray Matter  
C = Cerebrospinal fluid

**For Ch5 = L, R, N:**

**Character 6: Method Used**

H = Harvard Oxford Subcortical Atlas  
I = IIT Atlas v4.1 (used in MR1 only)  
K = IIT Atlas v5.0  
J = JHU Atlas

**Characters 7 & 8:**

Numbers = See table in

*M3\_P5\_DOCUMENTATION\_OF\_BRAIN\_MEASURES\_20260910*  
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